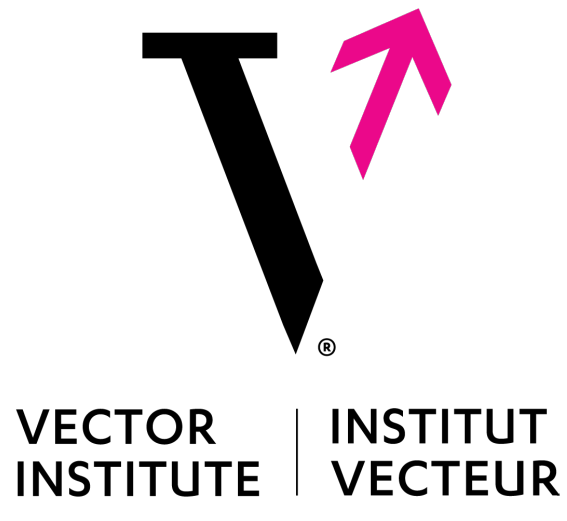




MDM-Prime-v2: Binary Encoding and Index Shuffling

Enable Scaling of Diffusion Language Models

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TL;DR

- MDM-Prime [1]: diffuse on **sub-tokens** → finer-grained denoising
- Issue**: base- b sub-tokens of BPE token IDs have **low entropy** → higher training loss
- Fix**: **Index Shuffling** + **Binary Encoding** via lookup tables

Zero-shot Accuracy

49.42

avg. zero-shot acc.
(8 tasks, best @1.1B)

+4.87

points over SMDM
(same setup, 1.1B)

Github Repository



MDM-Prime

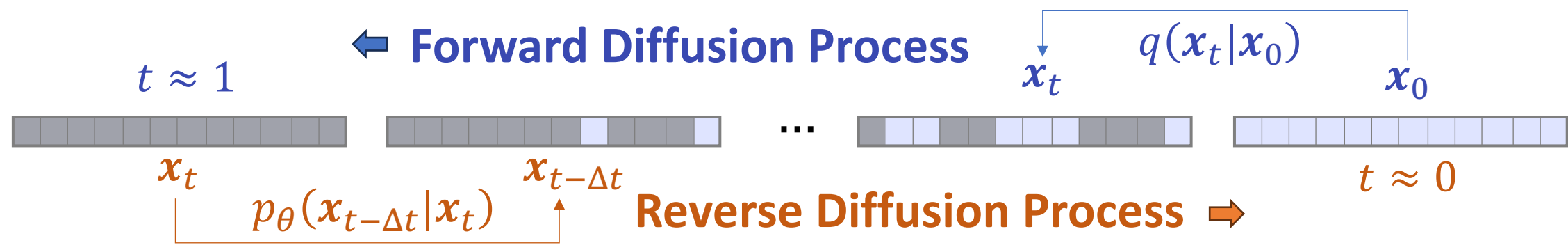


MDM-Prime-v2

Background

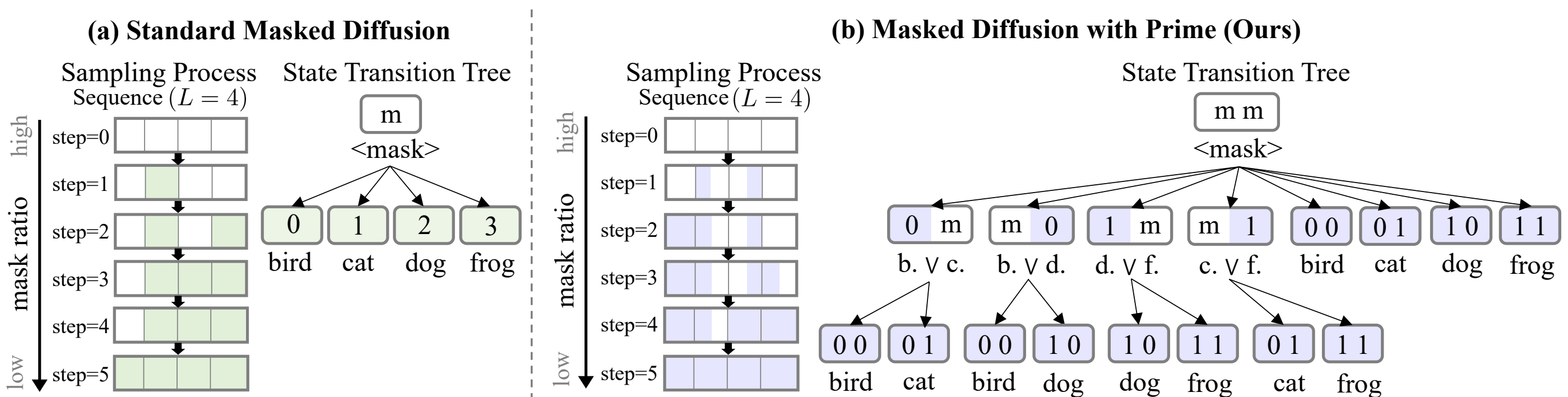
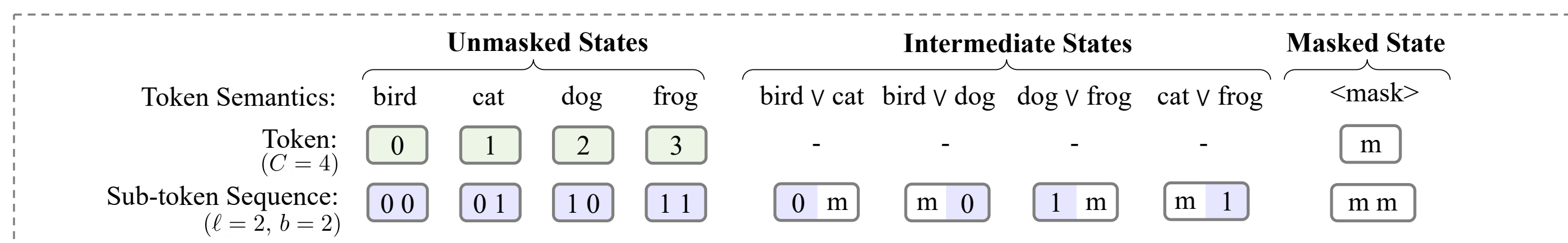
MDM-Prime Framework

- Masked Diffusion Models (MDM)

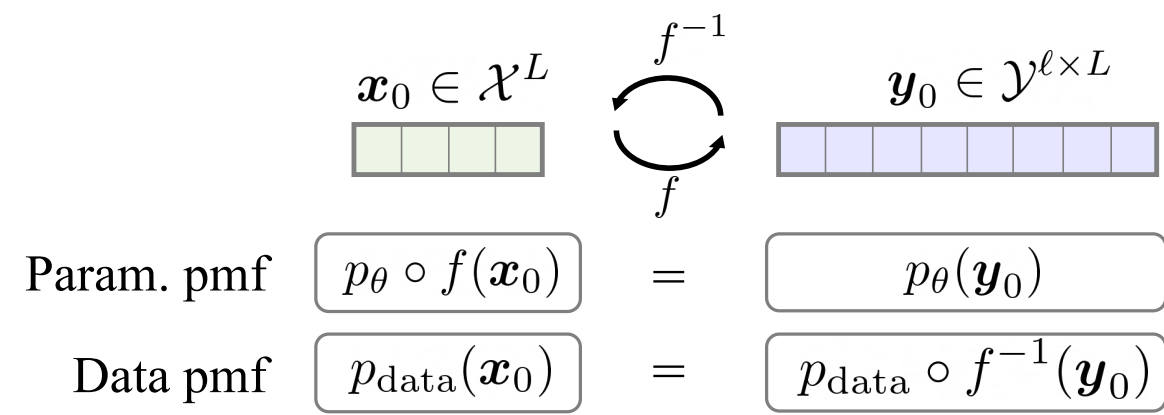


- MDM-Prime introduces intermediate states

→ alleviates idle step issue → better model utilization



- Invertible Base- b Encoding



- Cross-entropy Objective

$$\mathcal{L}^{(\ell)} = \int_0^1 \frac{\alpha'_t}{1 - \alpha_t} \mathbb{E}_{q(y_t|y_0)} \left[\sum_{i=1}^L \log p_\theta(y_0^i | y_t) \right] dt,$$

where $p_\theta(y_0^{i,j} | y_t) = 1$ for unmasked elements.

Experiments

Scaling Analysis

